LimeWire

v1.0 Studio Edition

Operation Manual

The modern music utility for everything.

18 tabs. 40+ integrations. One app.

Built with Python + tkinter + yt-dlp + Demucs + librosa + pydub

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Getting Started

LimeWire v1.0 Studio Edition is an all-in-one audio production toolkit with 18 specialized pages covering download, playback, analysis, editing, recording, stem separation, remixing, batch processing, and more.

SYSTEM REQUIREMENTS:
- Python 3.10 or newer
- Windows 10/11 (primary), macOS/Linux (partial support)
- FFmpeg installed and on PATH
- 4GB+ RAM recommended (8GB+ for Demucs stem separation)

QUICK START:
1. Install dependencies: pip install yt-dlp pillow requests mutagen pyglet
2. Install FFmpeg: winget install ffmpeg
3. Run: python LimeWire.py
4. Paste a URL and click 'Grab' to download your first track!

OPTIONAL MODULES (install as needed):
- Audio analysis: pip install librosa soundfile pyloudnorm
- Track ID: pip install shazamio musicbrainzngs pyacoustid
- Stem separation: pip install demucs
- Audio editing: pip install pydub sounddevice pyrubberband
- Transcription: pip install openai-whisper

Installation

CORE (required):

```
pip install yt-dlp pillow requests mutagen pyglet
winget install ffmpeg
```

ANALYSIS & IDENTIFICATION:

```
pip install librosa soundfile pyloudnorm
pip install musicbrainzngs pyacoustid
pip install shazamio   (Python 3.12 or earlier only)
```

STEM SEPARATION:

```
pip install demucs
(Includes PyTorch - GPU recommended for speed)
```

AUDIO EDITING & RECORDING:

```
pip install pydub sounddevice
pip install pyrubberband
pip install openai-whisper   (for transcription)
```

EFFECTS:

```
pip install pedalboard
```

DJ INTEGRATION:

```
pip install pyflp   (FL Studio project files)
```

DRAG & DROP (optional):

```
pip install tkinterdnd2
```

VERIFY INSTALLATION:

Run the app and check the status bar module counter (bottom-right).
It shows X/14 modules loaded. Click it to see which are missing.

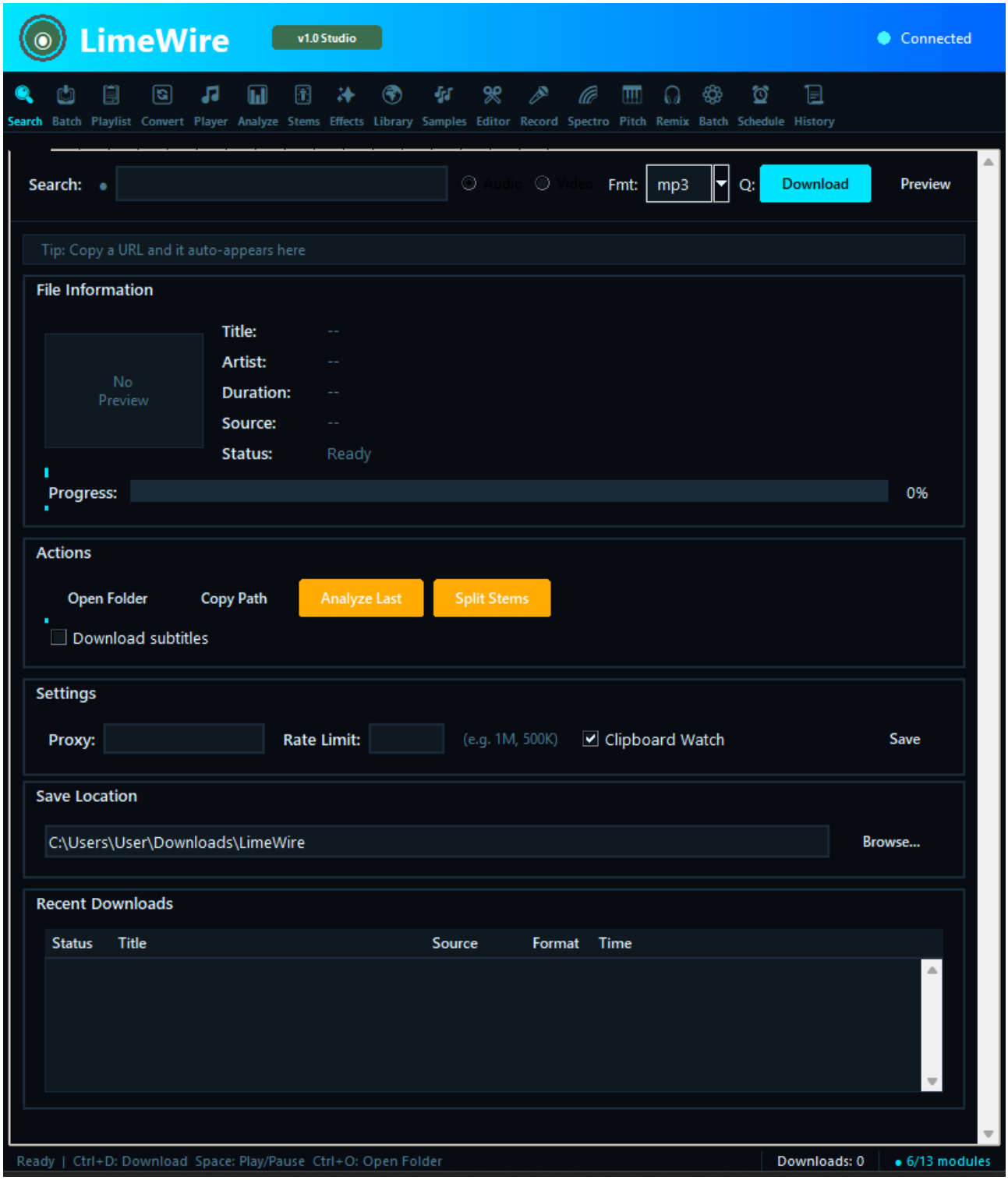
Keyboard Shortcuts

`Ctrl+D`	Download / Grab current URL
`Ctrl+O`	Open downloads folder
`Space`	Play / Pause audio
`Ctrl+K`	Open Command Palette (search pages & actions)
`Ctrl+?`	Show keyboard shortcuts dialog

Command Palette (Ctrl+K)

Press Ctrl+K anywhere in the app to open the Command Palette.
Type to fuzzy-search through all 18 pages and common actions.
Use arrow keys to navigate, Enter to select, Escape to dismiss.
Actions include: navigate to any page, open folder, cycle theme.

1. Search & Grab



The main entry point for downloading audio. Paste any URL from YouTube, Spotify, SoundCloud, Bandcamp, or 1000+ supported sites.

HOW TO USE:

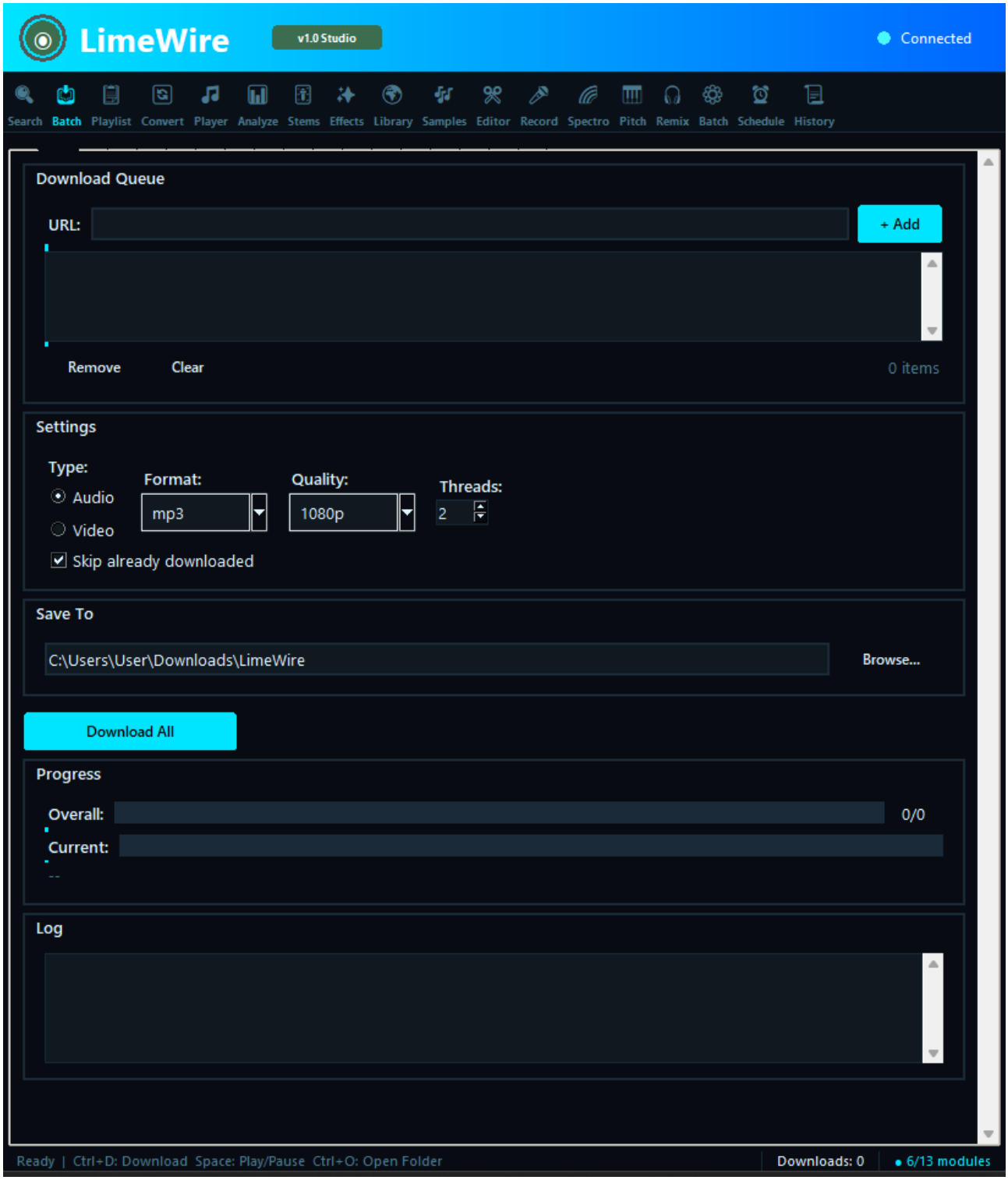
- Paste a URL into the URL field (auto-detected from clipboard)
- Select audio format: MP3, WAV, FLAC, OGG, M4A, AAC, OPUS
- Choose quality: Best, 320k, 256k, 192k, 128k
- Click 'Grab' or press Ctrl+D to start downloading

- Preview button fetches track info (title, artist, duration) before download

FEATURES:

- Auto-clipboard monitoring detects URLs as you copy them
- Spotify/Apple Music URLs are auto-resolved to YouTube for download
- Source auto-detection (YouTube, Spotify, SoundCloud, Bandcamp, etc.)
- Progress bar with speed indicator
- Downloads go to ~/Downloads/LimeWire/ by default

2. Batch Download



Download multiple URLs at once with a queue system.

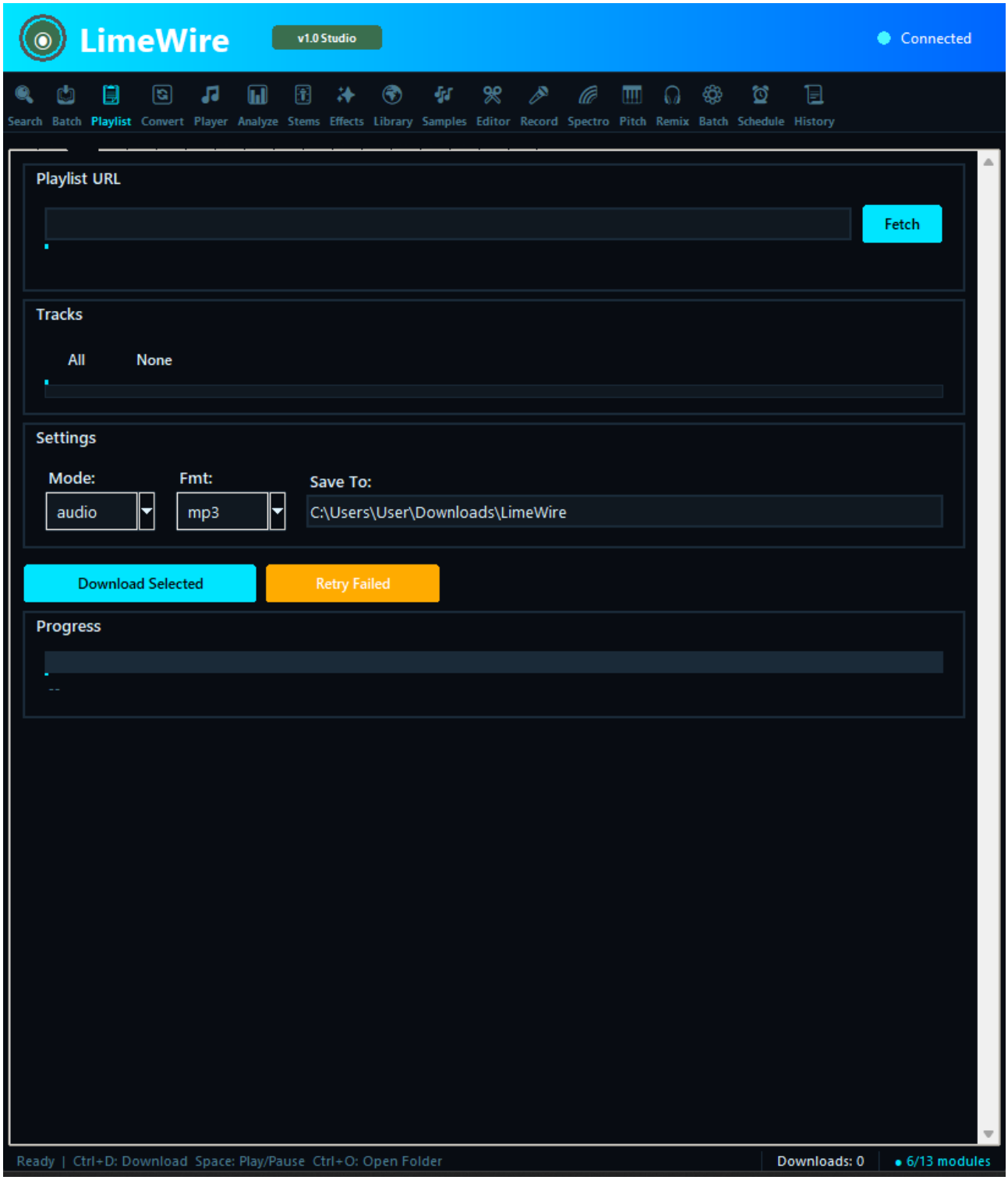
HOW TO USE:

- Enter one URL per line in the text area
- Click 'Add to Queue' to queue all URLs
- Click 'Start All' to begin downloading the entire queue
- Queue shows status: Pending, Downloading, Done, Error

FEATURES:

- Persistent queue saved to ~/.limewire_queue.json
- Individual retry for failed downloads
- Clear completed items
- Same format/quality options as Search & Grab

3. Playlist Download



Download entire YouTube playlists or channel content.

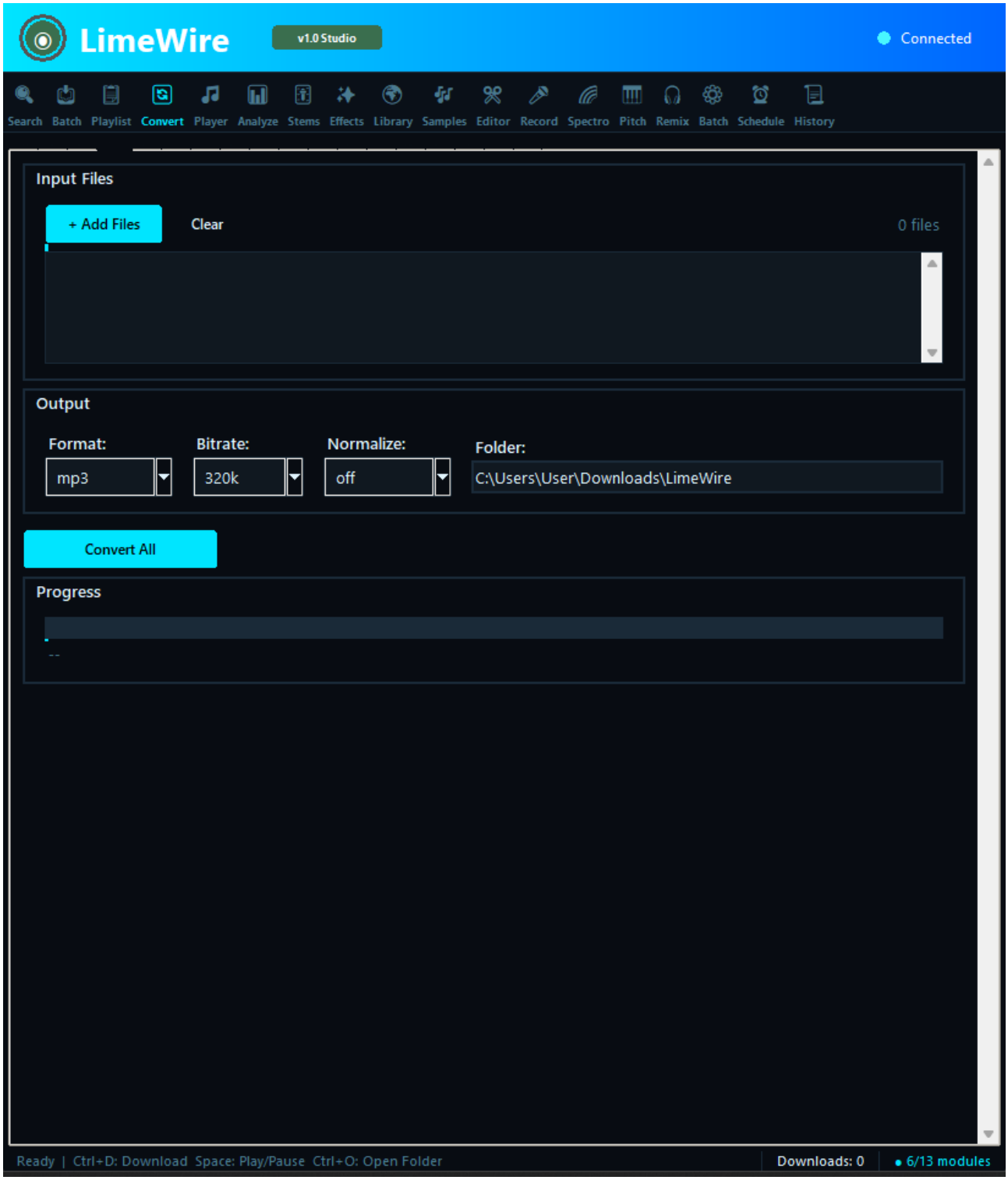
HOW TO USE:

- Paste a playlist URL
- Click 'Fetch Playlist' to see all tracks
- Select specific tracks or use 'Select All'
- Click 'Download Selected' to start batch download

FEATURES:

- Shows playlist title, track count, total duration
- Individual track selection with checkboxes
- Progress tracking per track

4. Converter



Convert audio files between formats with ffmpeg.

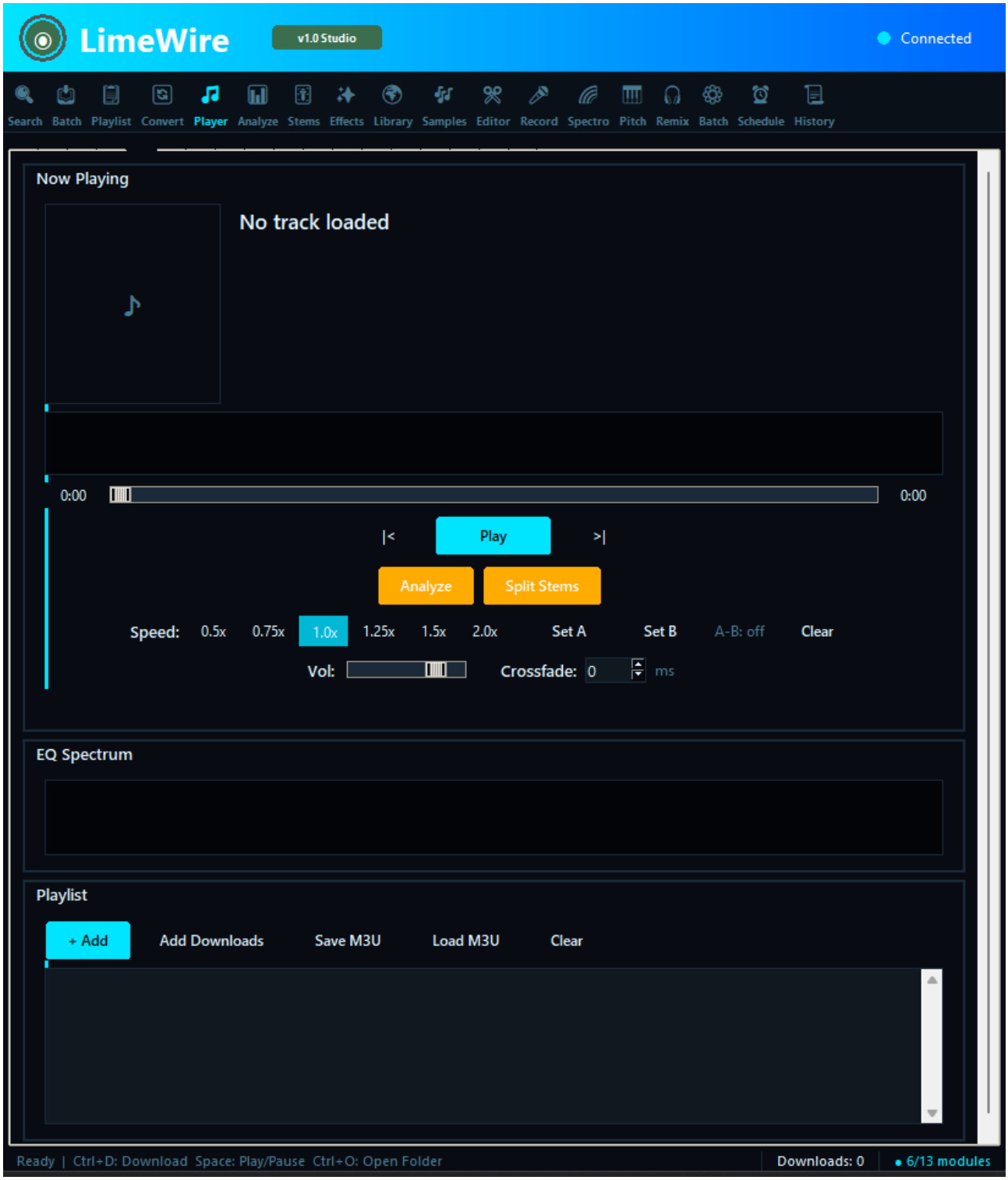
HOW TO USE:

- Browse for an audio file or drag & drop
- Select output format (MP3, WAV, FLAC, OGG, M4A, AAC, OPUS)
- Choose bitrate for lossy formats
- Click 'Convert'

FEATURES:

- Supports all major audio formats
- Preserves metadata during conversion
- Batch conversion via Batch Processor page

5. Player



Full-featured audio player with waveform visualization and EQ spectrum.

HOW TO USE:

- Add files via '+ Add' button or 'Add Downloads' for your library
- Double-click a track in the playlist to play
- Click the waveform to seek to any position
- Use speed controls to change playback rate (0.5x - 2.0x)
- Set A-B loop points for section repeat

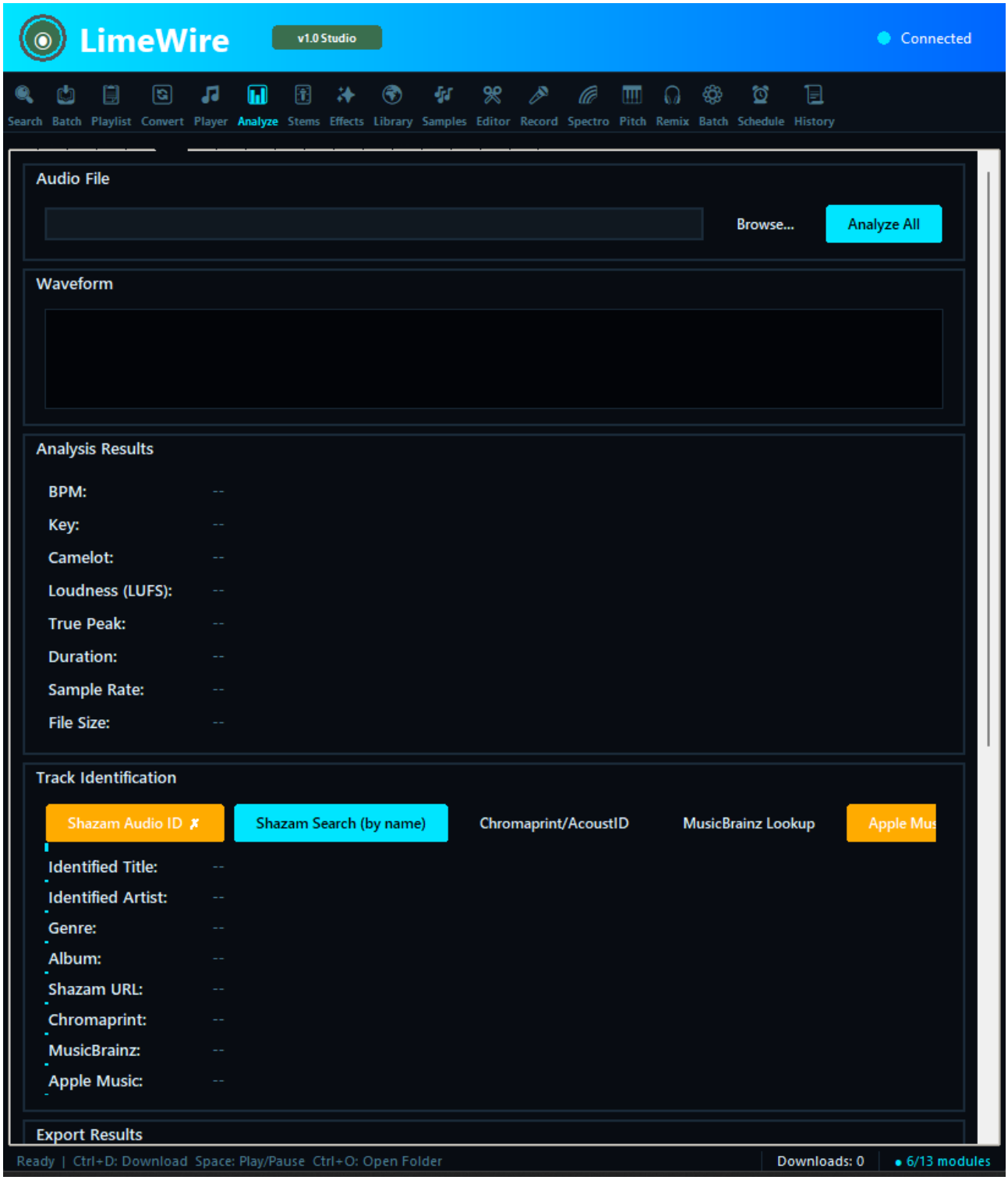
FEATURES:

- Waveform display with color-coded amplitude (green/yellow/orange/red)
- Real-time EQ spectrum visualizer (32-band)
- Album art extraction from ID3/FLAC/OGG tags
- Crossfade control (0-5000ms) between tracks
- 'Up Next' indicator shows the next queued track
- Volume control with speed adjustment
- M3U playlist save/load
- Quick access: 'Analyze' and 'Split Stems' buttons for current track

KEYBOARD SHORTCUTS:

- Space: Play/Pause
- Arrow keys in waveform: Seek

6. Analyze



Comprehensive audio analysis: BPM, key, loudness, and track identification.

HOW TO USE:

- Browse for an audio file
- Click 'Analyze All' to run full analysis
- Use identification buttons for Shazam, MusicBrainz, Chromaprint, Apple Music
- Use 'Write Tags to File' to save identified metadata

ANALYSIS RESULTS:

- BPM (beats per minute) via librosa
- Musical key (e.g., C major, A minor)
- Camelot wheel notation for DJ mixing
- Loudness in LUFS (integrated) and true peak
- Duration, sample rate, file size
- Waveform visualization

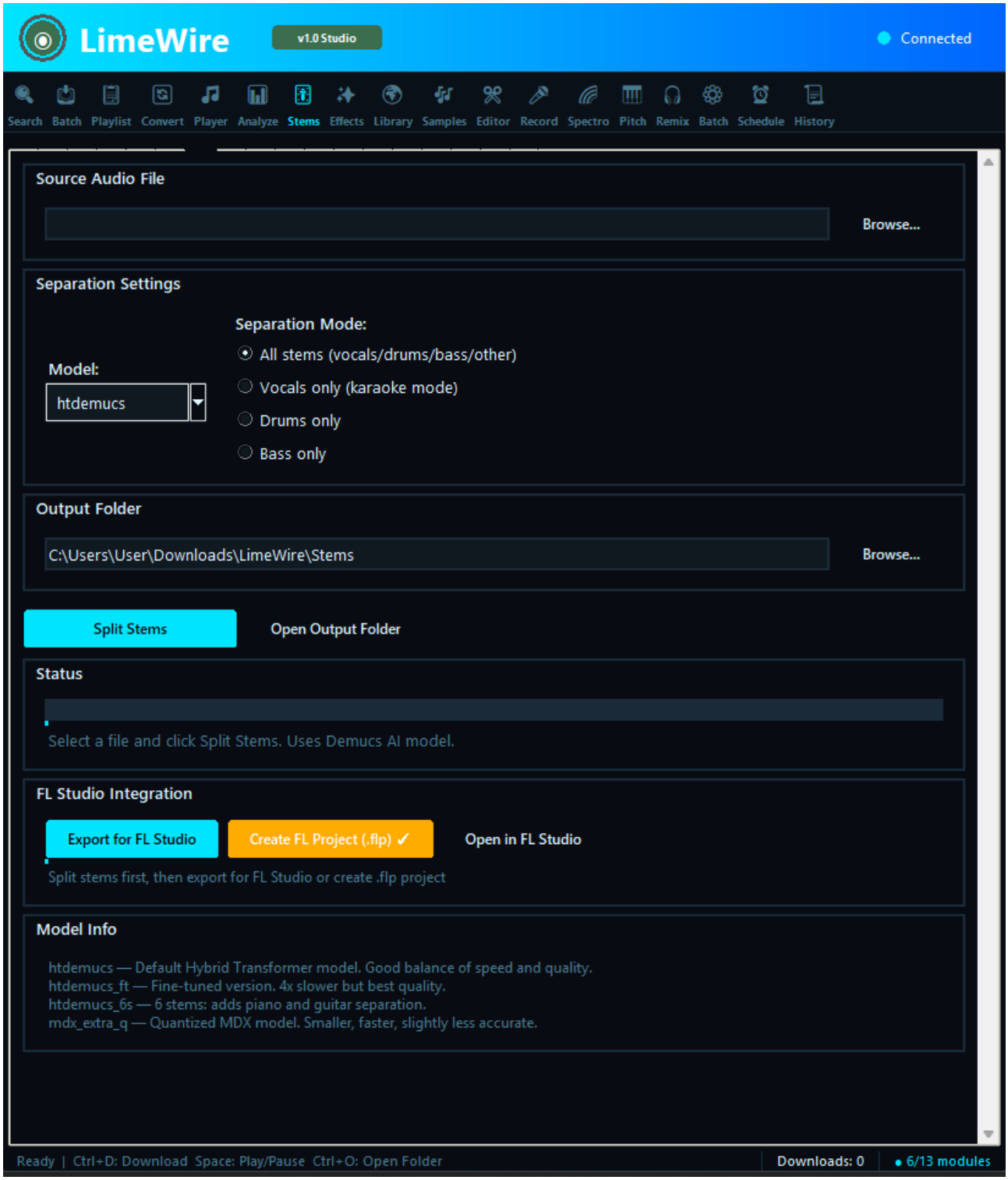
LOUDNESS TARGETING:

- Select a platform preset: Spotify (-14 LUFS), YouTube (-13), Apple Music (-16), CD/Master (-9), Club (-6), Podcast (-16)
- Click 'Normalize to Target' for 2-pass ffmpeg loudnorm
- Shows before/after LUFS measurements

AUDIO TOOLS:

- Noise reduction, lyrics lookup, effects chain
- DJ Integration: Serato tags, FL Studio opening

7. Stems (Demucs)



Separate audio into individual stems using AI (Demucs by Meta).

HOW TO USE:

- Browse for an audio file
- Select model: htdemucs (fast), htdemucs_ft (quality), mdx_extra
- Choose separation mode: All Stems, Vocals Only, Instrumental
- Click 'Separate' and wait (may take several minutes)

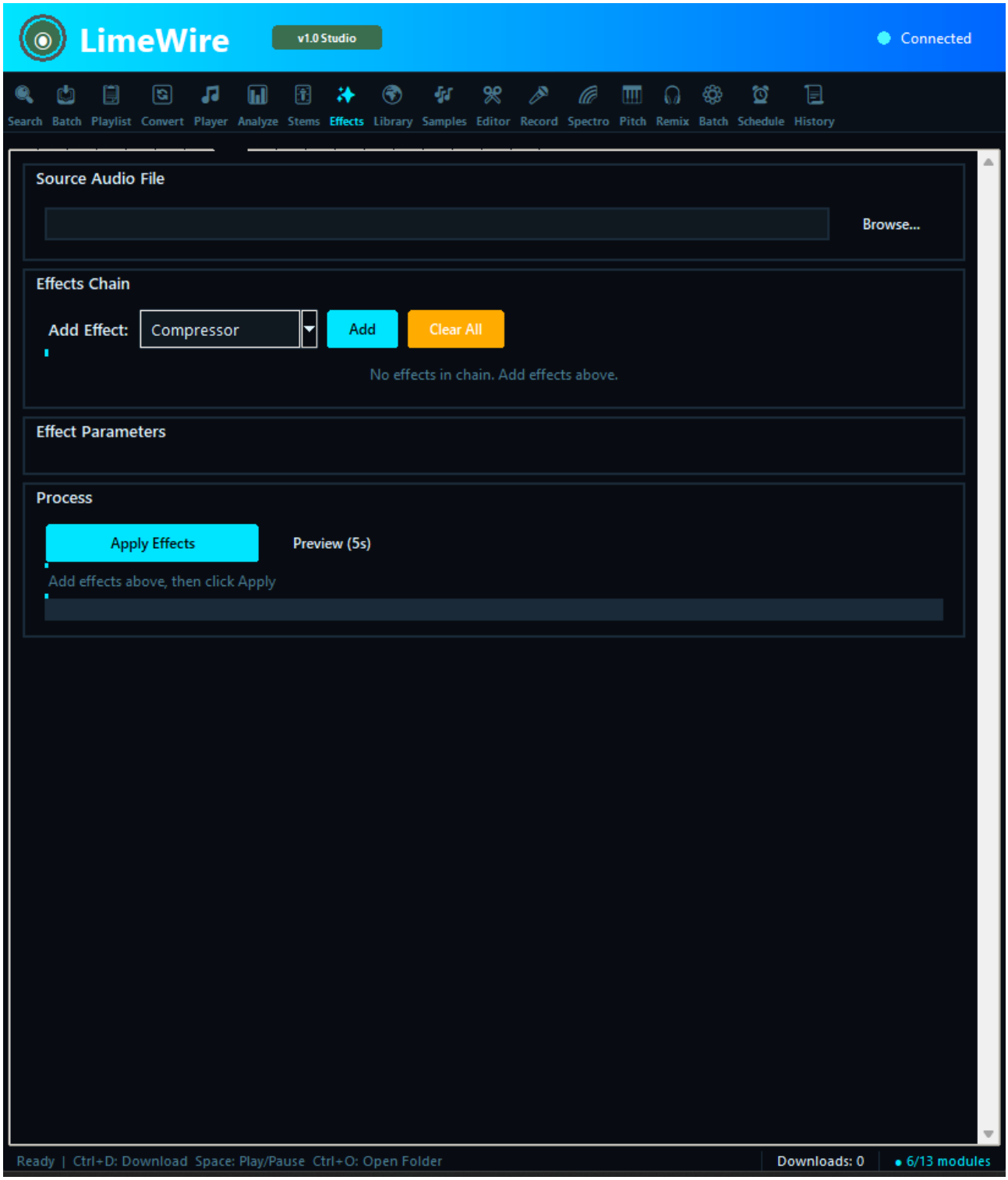
OUTPUT STEMS:

- Vocals, Drums, Bass, Other (4-stem models)
- Piano, Guitar (6-stem models like htdemucs_6s)
- Output saved to Downloads/LimeWire/Stems/

REQUIREMENTS:

- pip install demucs (includes PyTorch)
- GPU recommended for faster processing

8. Effects



Apply audio effects via pedalboard (Spotify's audio plugin framework).

HOW TO USE:

- Browse for an audio file
- Enable desired effects and adjust parameters
- Click 'Apply Effects' to process

AVAILABLE EFFECTS:

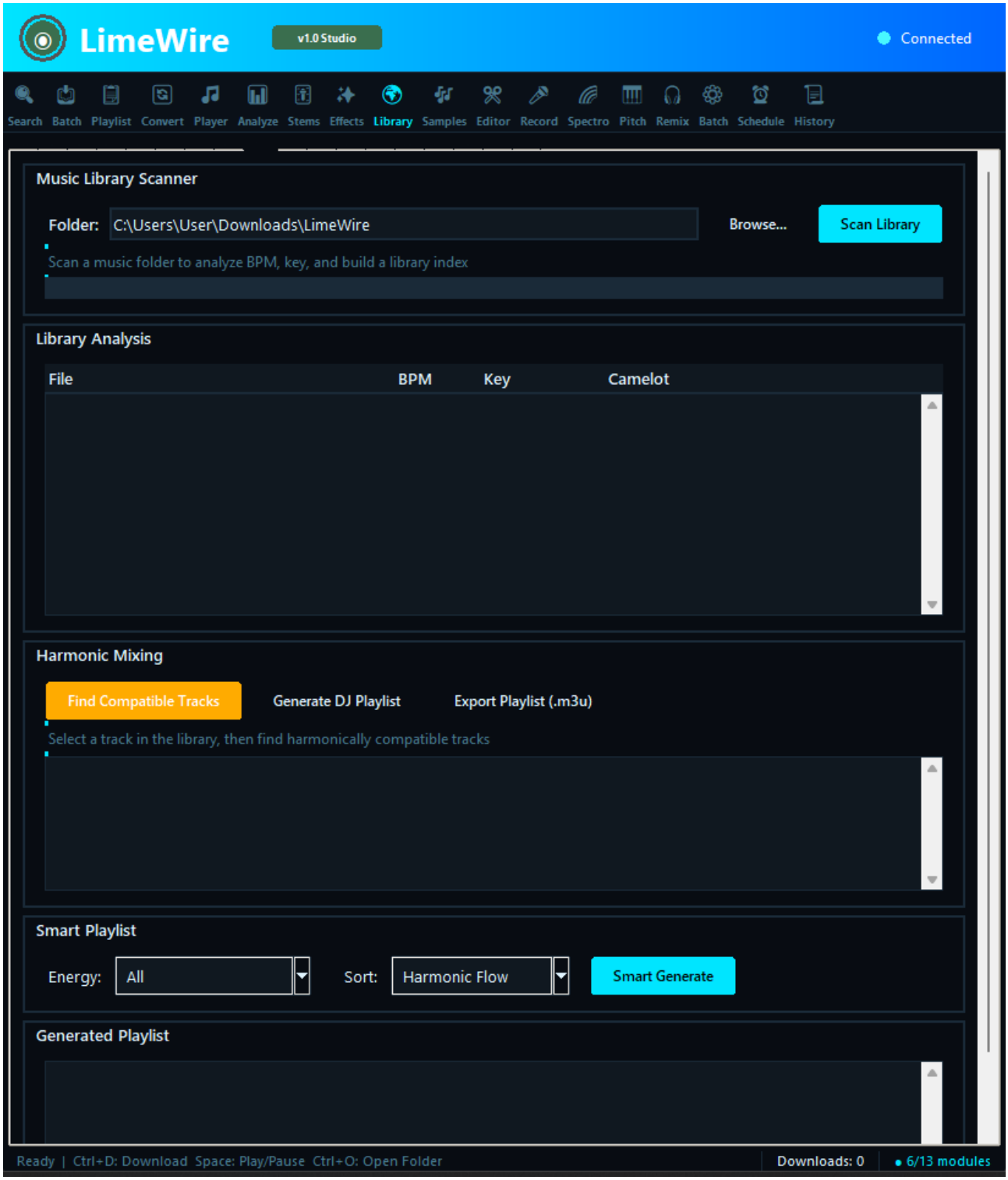
- Gain, Compressor, Limiter

- Reverb, Delay, Chorus
- High-pass / Low-pass filters
- EQ adjustments

REQUIREMENTS:

- pip install pedalboard

9. Discovery



Music library scanner with BPM/key analysis and harmonic mixing tools.

HOW TO USE:

- Set your music folder path
- Click 'Scan Library' to analyze all audio files
- Select a track to find harmonically compatible songs
- Generate DJ playlists with harmonic progression

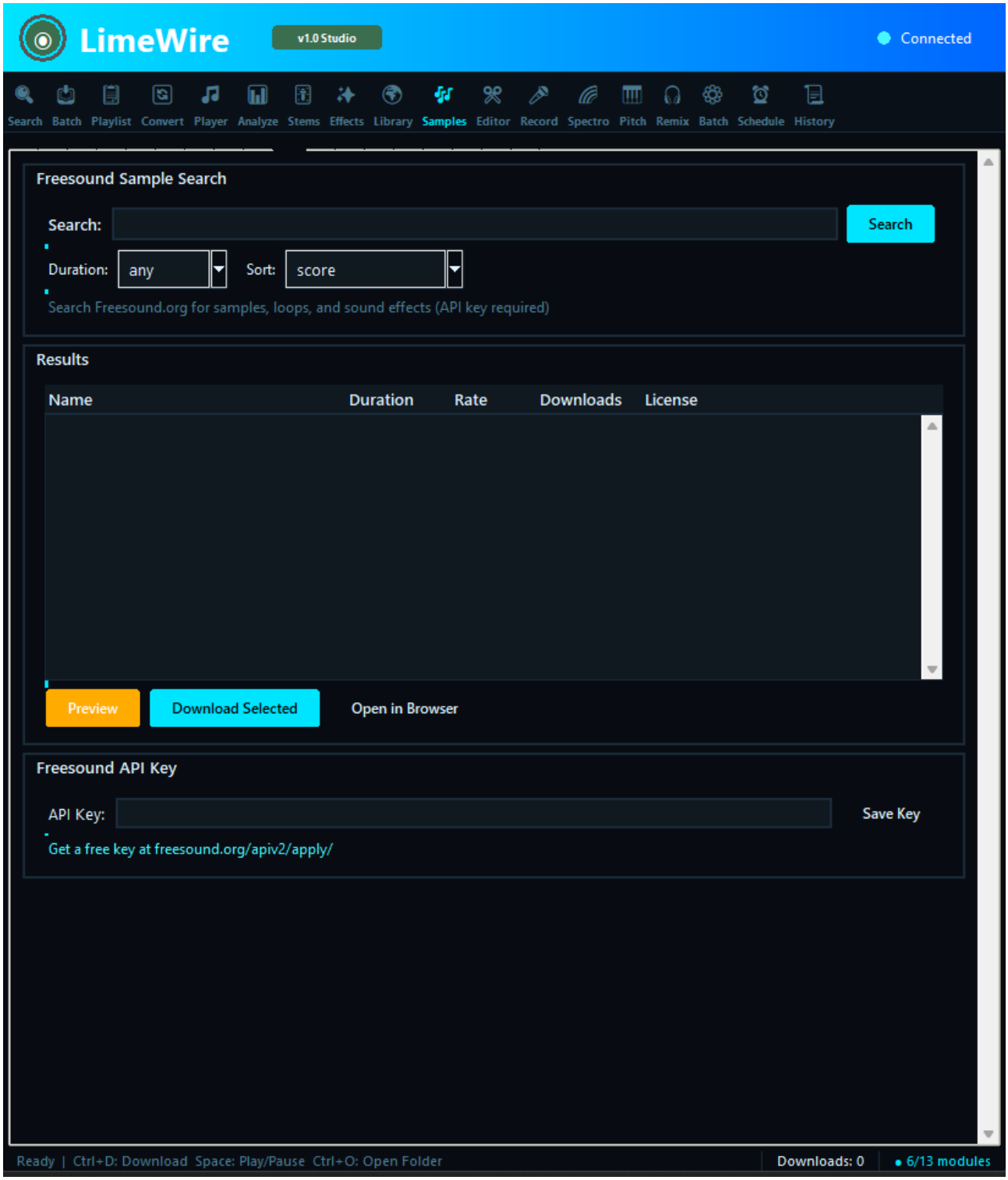
SMART PLAYLISTS:

- Energy filter: Low (<100 BPM), Medium (100-130), High (130-160), Very High (160+)
- Sort modes: Harmonic Flow, BPM Ramp Up, BPM Ramp Down, Key Groups
- Click 'Smart Generate' for filtered playlist creation
- Export as .m3u with BPM/key metadata

FEATURES:

- Cached analysis results (~/.limewire_analysis_cache.json)
- Camelot wheel compatibility detection
- Harmonic key matching for smooth DJ transitions

10. Samples



Browse and download free samples from Freesound.org.

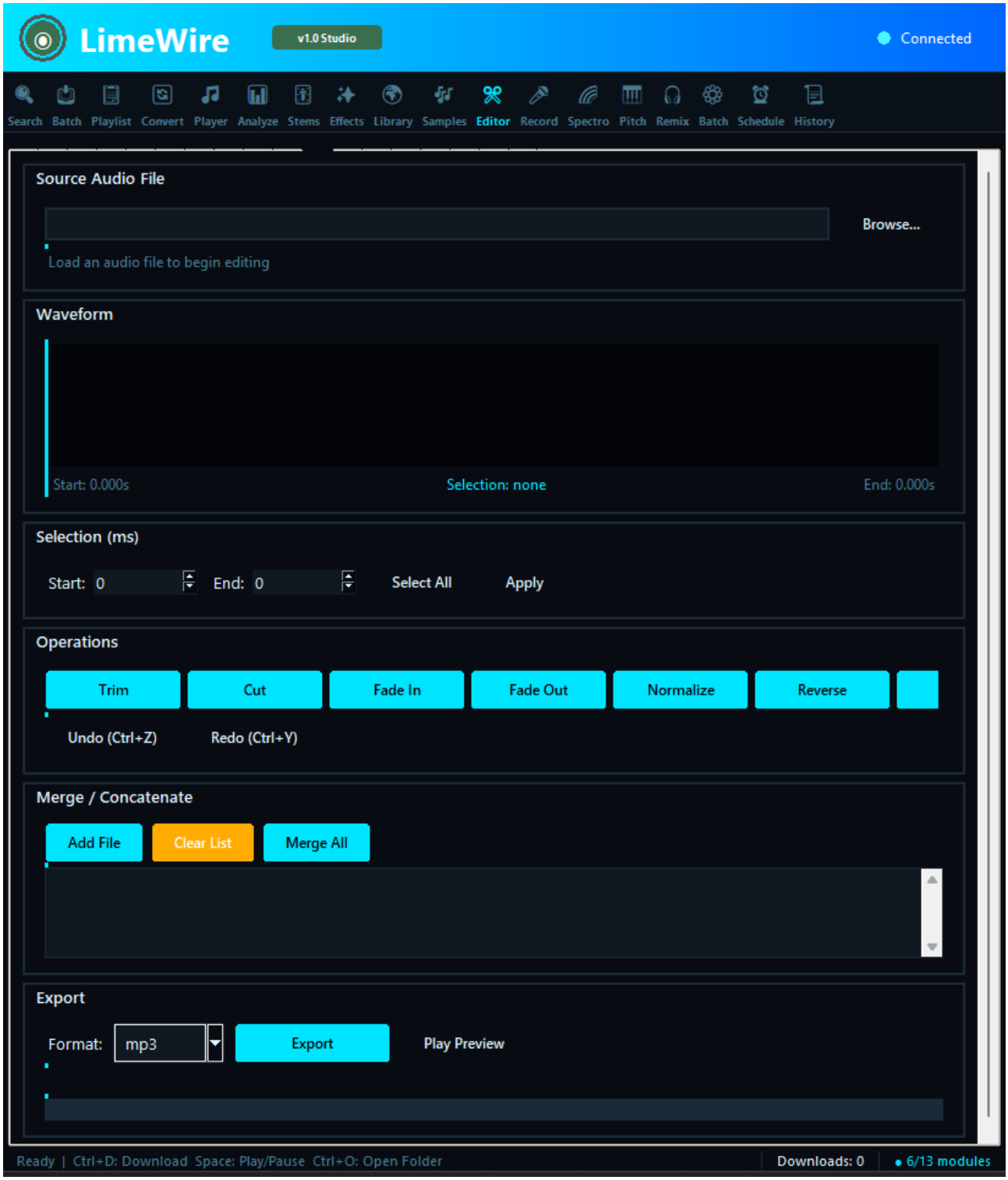
HOW TO USE:

- Enter a search query (e.g., 'kick drum', 'ambient pad')
- Click 'Search' to find matching samples
- Preview samples directly in the app
- Download selected samples to your library

NOTE:

- Requires a Freesound API key (free registration)
- Samples are CC-licensed for use in your projects

11. Editor



Non-destructive audio editor with undo/redo support.

HOW TO USE:

- Load an audio file
- Use waveform display to select regions (click and drag)
- Apply operations: Trim, Cut, Fade In/Out, Merge
- Undo/Redo with Ctrl+Z / Ctrl+Y
- Export the edited result

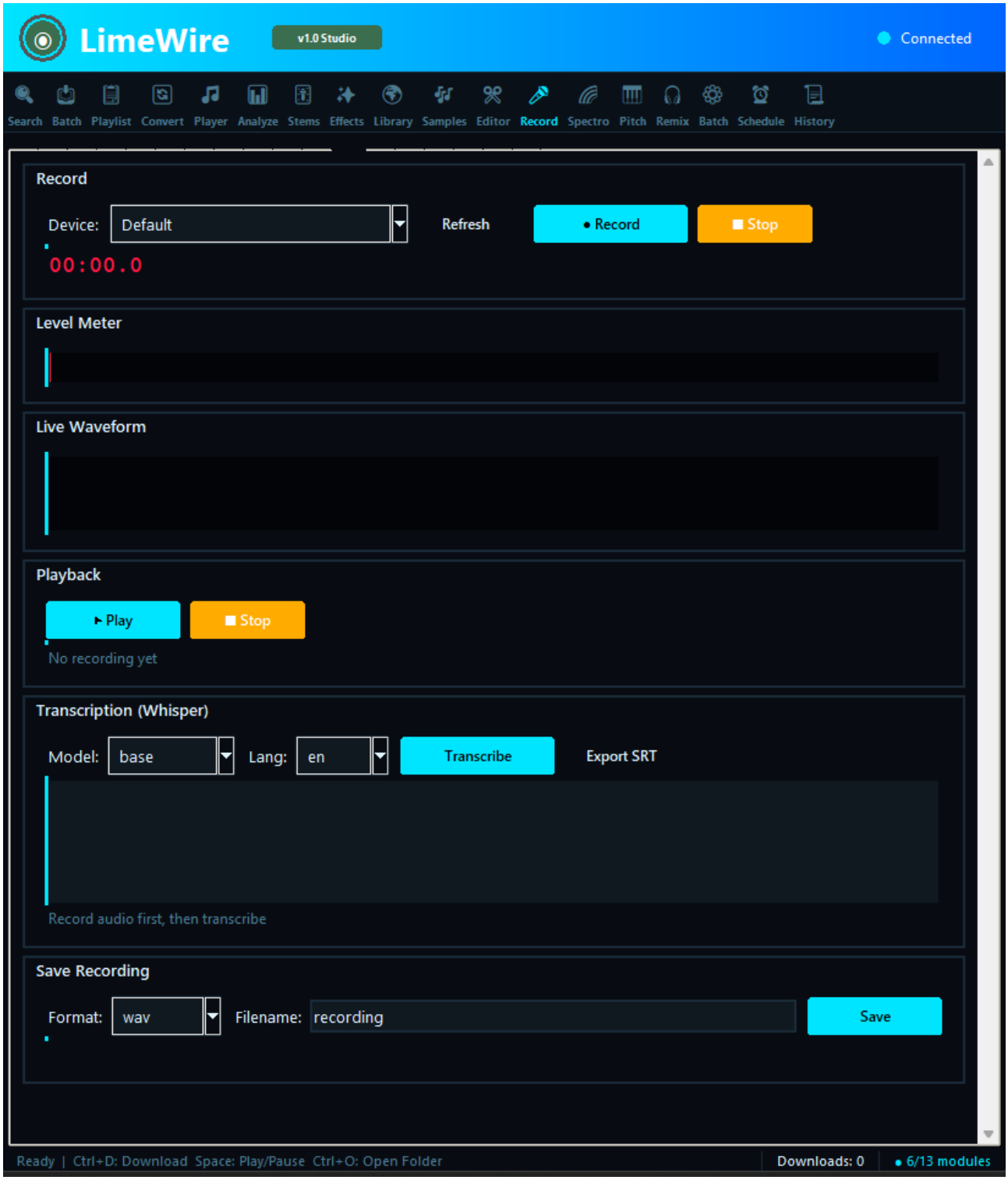
OPERATIONS:

- Trim: Keep only the selected region
- Cut: Remove the selected region
- Fade In: Apply fade-in to selection start
- Fade Out: Apply fade-out to selection end
- Merge: Append another audio file

REQUIREMENTS:

- pip install pydub

12. Recorder



Record audio from your microphone with real-time monitoring.

HOW TO USE:

- Select input device from dropdown
- Click 'Record' to start capturing
- Watch the VU meter and live waveform
- Click 'Stop' when done
- Optionally transcribe with Whisper AI

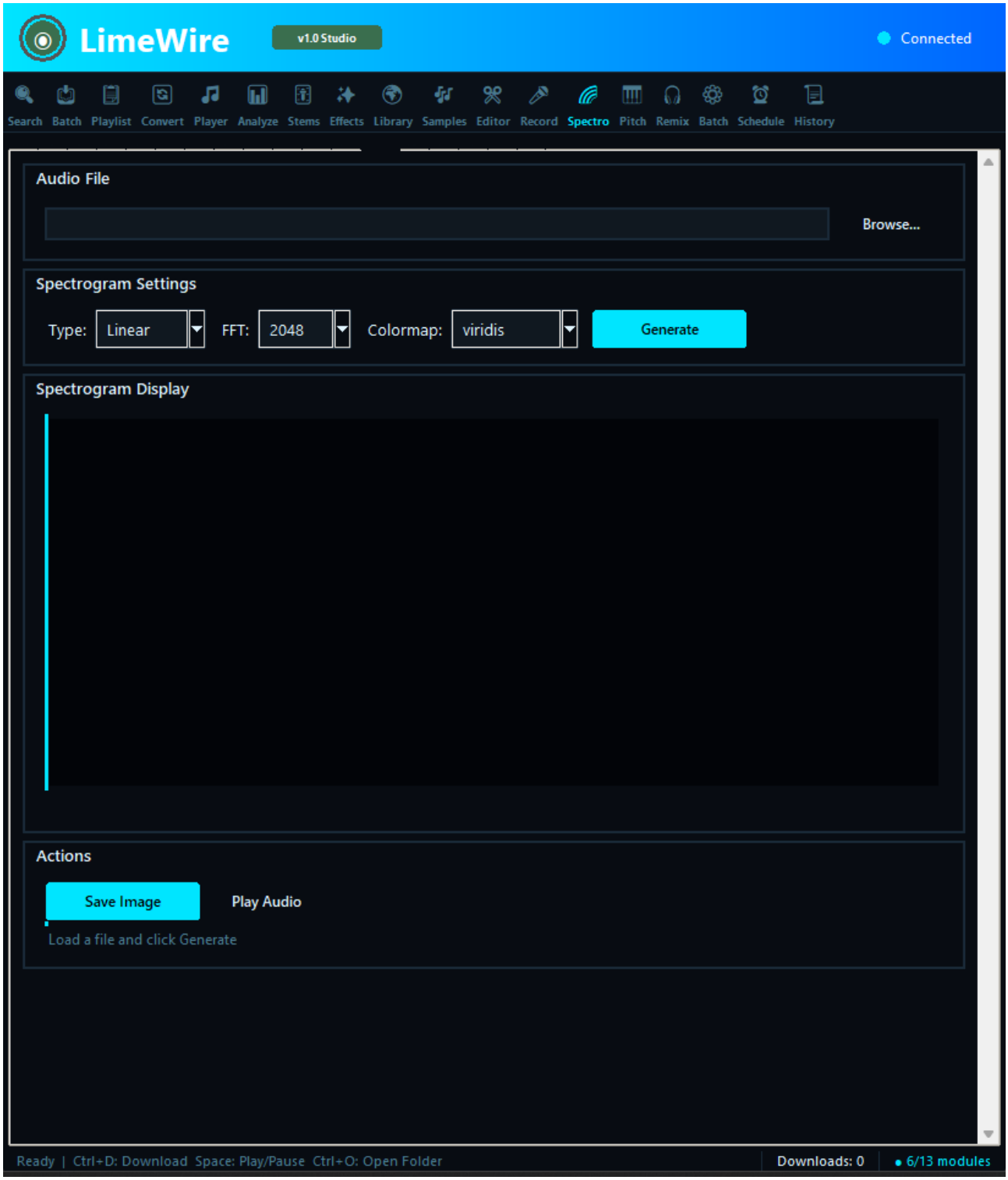
FEATURES:

- Real-time VU meter (level monitoring)
- Live waveform display during recording
- Whisper transcription with SRT subtitle export
- Save as WAV format

REQUIREMENTS:

- pip install sounddevice (recording)
- pip install openai-whisper (transcription, optional)

13. Spectrogram



Generate visual spectrogram representations of audio files.

HOW TO USE:

- Browse for an audio file
- Select spectrogram type: Linear, Mel, CQT (Constant-Q)
- Choose colormap: Viridis, Magma, Plasma, Inferno
- Click 'Generate' to create the spectrogram
- Export as PNG image

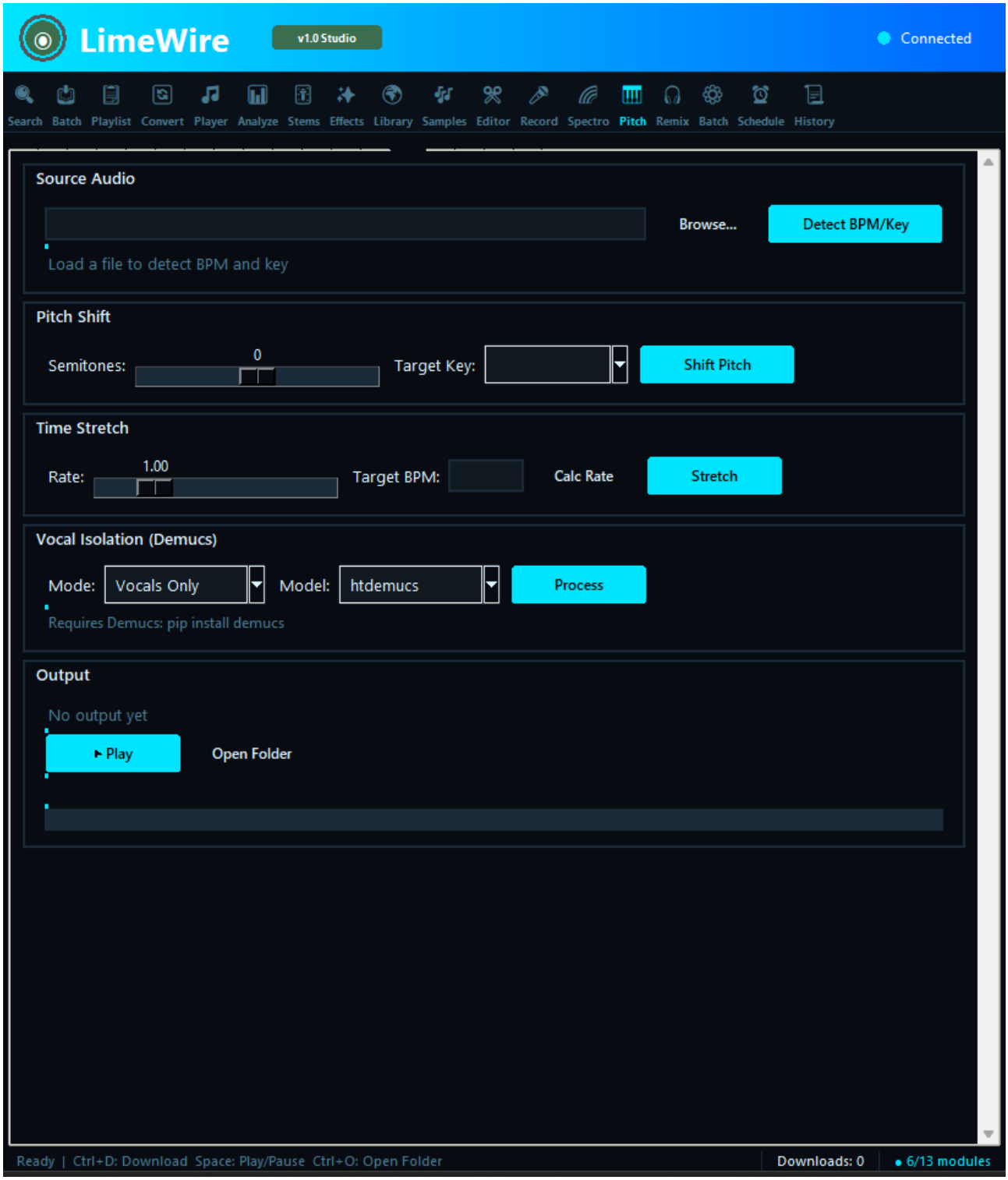
SPECTROGRAM TYPES:

- Linear: Standard frequency vs time display
- Mel: Perceptually-weighted frequency scale (matches human hearing)
- CQT: Constant-Q transform, ideal for musical analysis

REQUIREMENTS:

- pip install librosa matplotlib

14. Pitch/Time



Pitch shifting, time stretching, and vocal isolation tools.

HOW TO USE:

- Browse for an audio file
- Pitch Shift: Set semitones (+/- 12) and click 'Shift Pitch'
- Time Stretch: Set rate (0.5x - 2.0x) and click 'Stretch'
- Auto BPM: Detect current BPM, set target BPM, auto-calculate rate
- Vocal Isolation: Select mode and Demucs model, click 'Isolate'

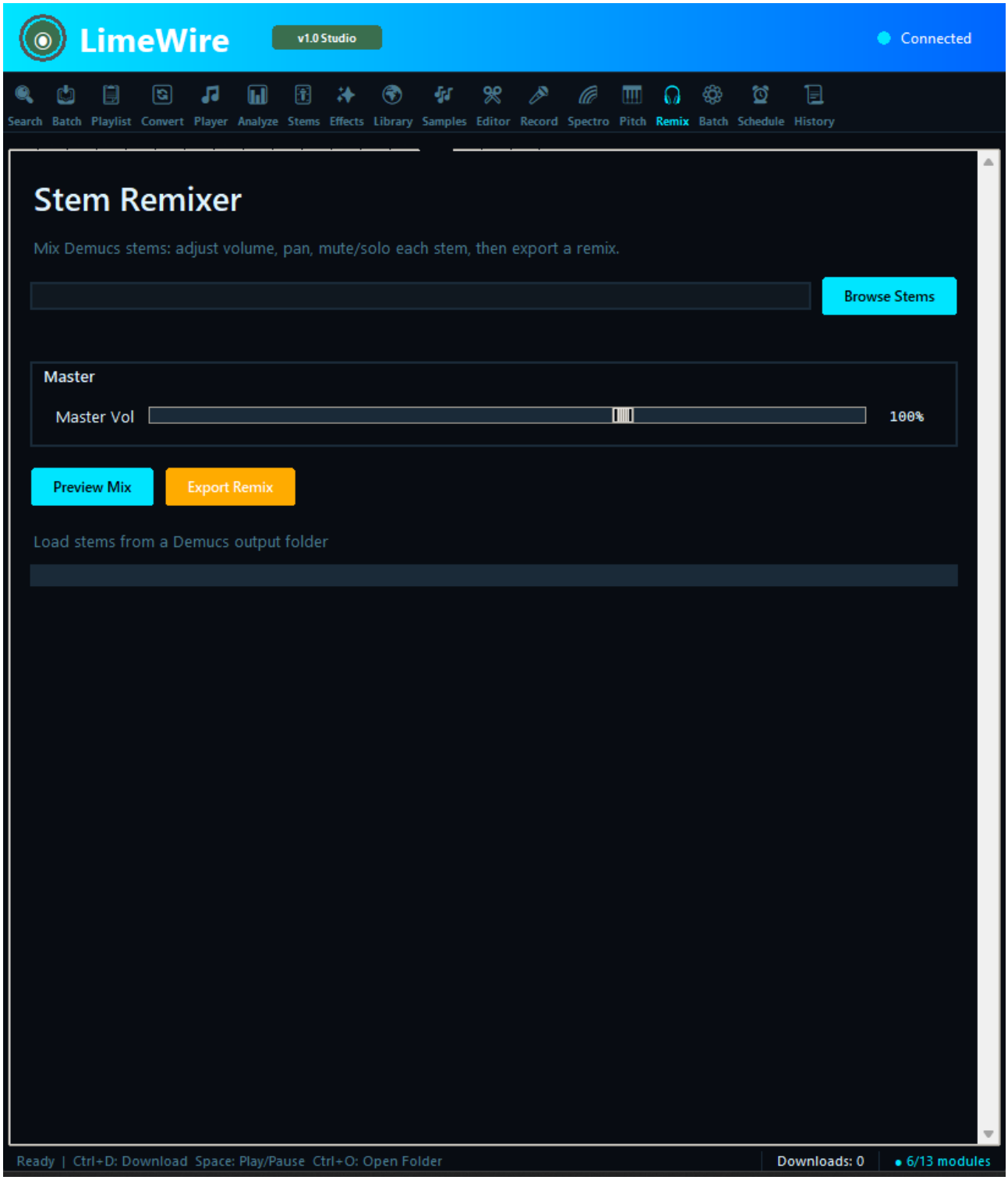
FEATURES:

- Preserves audio quality using pyrubberband (rubber band library)
- BPM auto-detection for precise tempo matching
- Multiple vocal isolation modes: Vocals Only, Instrumental, All Stems

REQUIREMENTS:

- pip install pyrubberband soundfile (pitch/time)
- pip install demucs (vocal isolation)

15. Remixer



Mix Demucs-separated stems with per-stem controls.

HOW TO USE:

- First separate a track using the Stems page
- Click 'Browse Stems' and select the Demucs output folder
- **Each stem gets its own channel strip with controls:**
  - Volume slider (0-150%)
  - Pan slider (L100 to R100, C = center)

  - Mute (M) checkbox
  - Solo (S) checkbox
- Adjust Master Volume (0-150%)
- Click 'Preview Mix' to hear your remix
- Click 'Export Remix' to save as WAV/MP3/FLAC

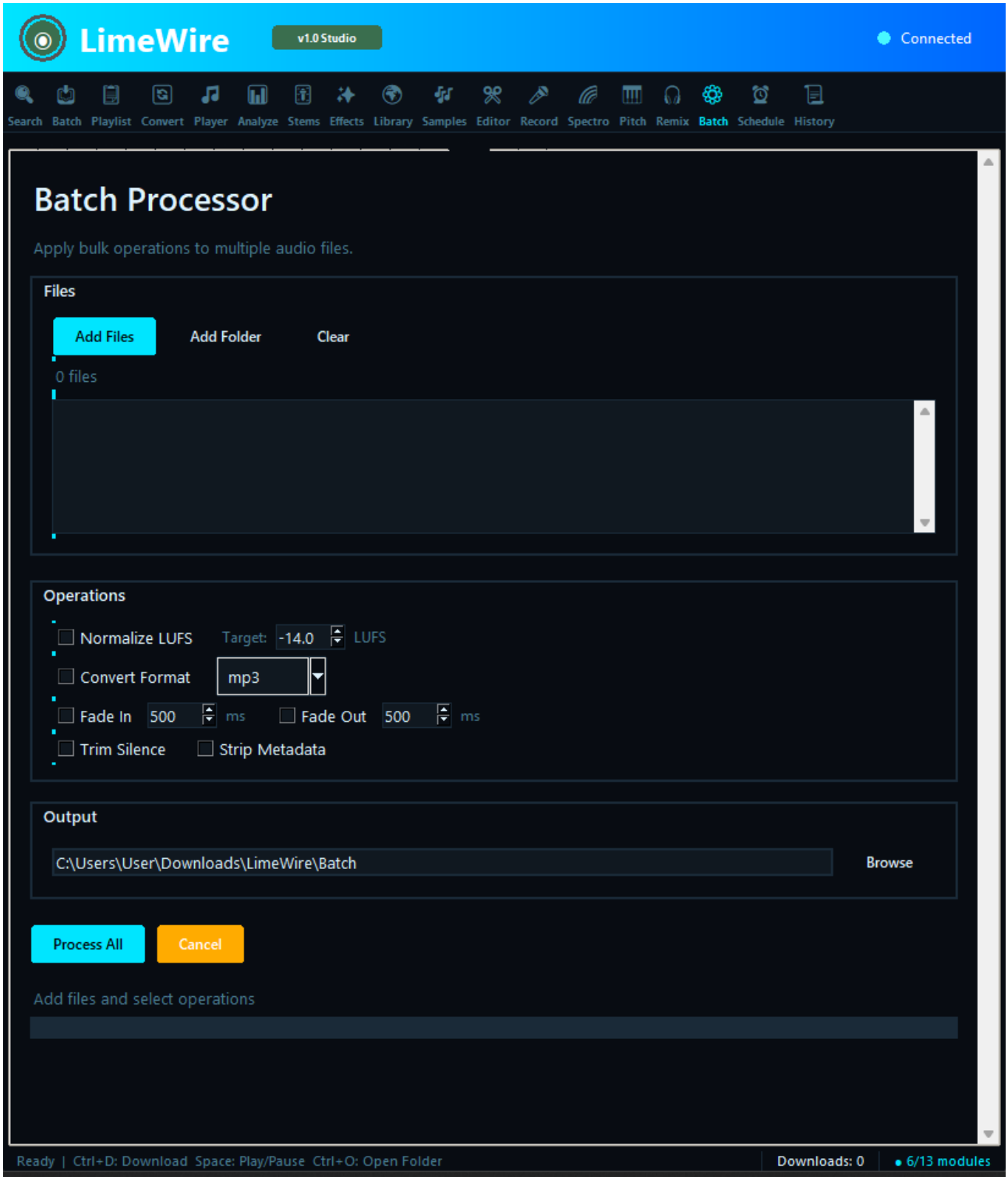
STEM COLOR CODING:

- Vocals = Green, Drums = Orange, Bass = Blue
- Other = Gray, Piano = Yellow, Guitar = Red

REQUIREMENTS:

- pip install pydub
- Demucs-separated stems (from Stems page)

16. Batch Processor



Apply bulk operations to multiple audio files at once.

HOW TO USE:

- Click 'Add Files' or 'Add Folder' to select audio files
- **Enable desired operations with checkboxes:**
  - Normalize LUFS: Target loudness level (e.g., -14 LUFS)
  - Convert Format: Output as MP3/WAV/FLAC/OGG/M4A
  - Fade In / Fade Out: Duration in milliseconds

  - Trim Silence: Remove leading/trailing silence
  - Strip Metadata: Remove all tags from output files
- Set output folder
- Click 'Process All' to start batch processing
- Cancel button stops processing

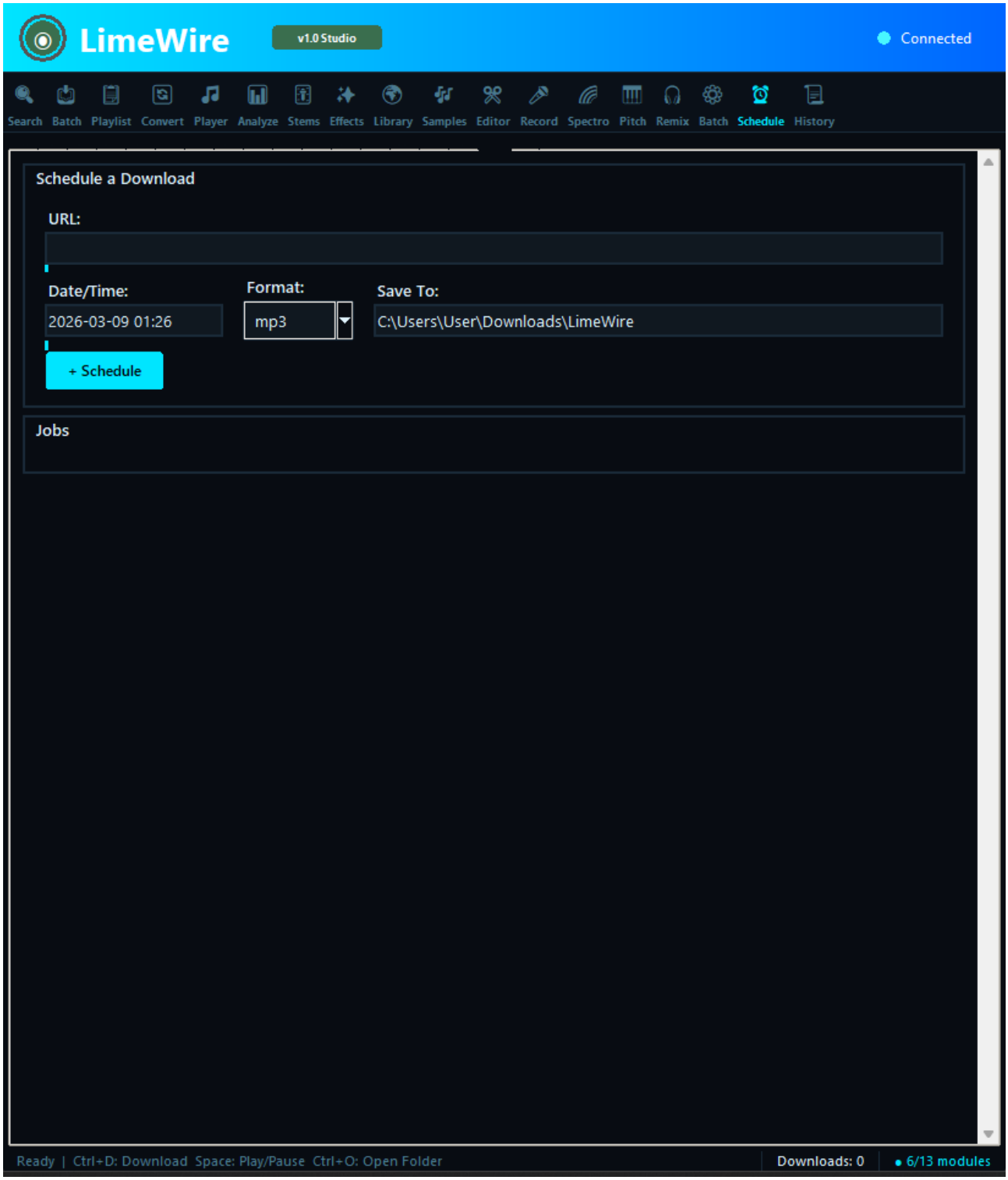
FEATURES:

- Per-file and overall progress tracking
- Error count reporting
- Cancellation support

REQUIREMENTS:

- pip install pydub

17. Scheduler



Schedule downloads for later or set up recurring downloads.

HOW TO USE:

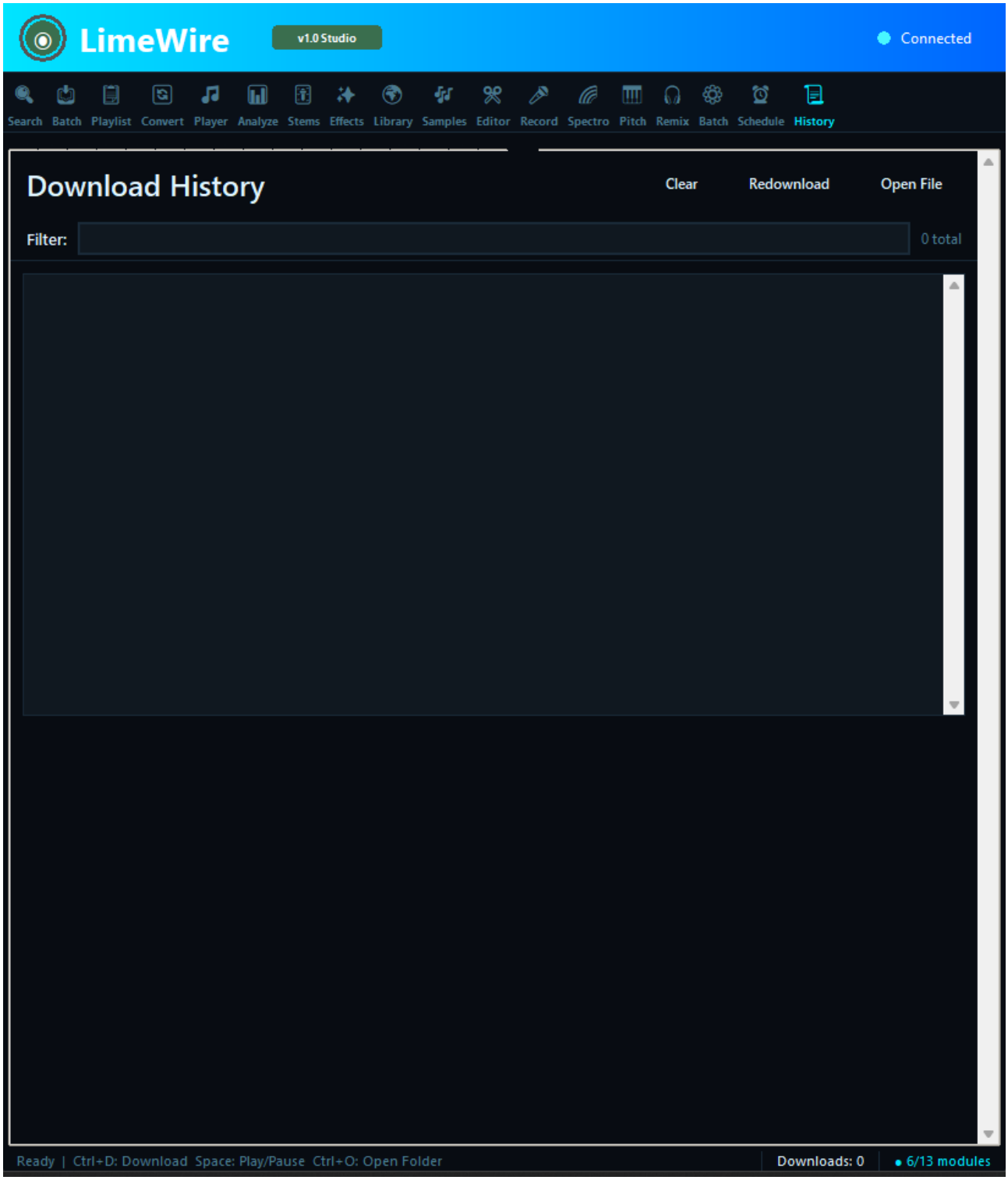
- Enter a URL and set the download time
- Click 'Schedule' to add to the queue
- The scheduler runs in the background and triggers at the set time

FEATURES:

- Persistent schedule saved to ~/.limewire_schedule.json

- Background polling every 30 seconds
- Status tracking: Pending, Running, Complete, Failed

18. History



Complete download history with search and management.

HOW TO USE:

- All downloads are automatically logged
- Search/filter by filename
- Double-click to play a track
- Right-click for options (open folder, re-download, delete)

FEATURES:

- Persistent history saved to ~/.limewire_history.json
- Sortable columns
- Clear history option in Tools menu

Theme Customization

LimeWire includes 3 built-in themes that switch live (no restart needed):

CLASSIC LIGHT:
Warm neutral background (#F8F9FA), soft green accents (#2ECC71).
Easy on the eyes for daytime use.

CLASSIC DARK:
Rich dark background (#1A1D21), green accents. Default theme.
Perfect for studio environments and low-light conditions.

MODERN DARK:
GitHub-inspired dark (#0D1117), brighter green (#3FB950).
High contrast for focused work sessions.

HOW TO SWITCH:
- Menu: Tools > Cycle Theme (Light/Dark/Modern)
- Command Palette: Ctrl+K, type 'theme'
- Theme preference is saved and restored on next launch

Troubleshooting

COMMON ISSUES:

'FFmpeg not found':

Install FFmpeg: winget install ffmpeg

Or download from https://ffmpeg.org and add to PATH.

'Module not found' errors:

Check the module counter in the status bar (bottom-right).

Install missing modules with pip. See Installation section.

Downloads fail or are slow:

Update yt-dlp: pip install -U yt-dlp

Check Tools > Check yt-dlp Update for version info.

Some sites may require cookies or authentication.

Demucs/stems taking too long:

GPU acceleration highly recommended for stem separation.

Install PyTorch with CUDA: pip install torch --index-url ...

The htdemucs model is fastest; htdemucs_ft is higher quality but slower.

Whisper transcription fails:

Ensure openai-whisper is installed: pip install openai-whisper

First run downloads the model (may take a few minutes).

Audio playback issues:

Ensure pyglet is installed: pip install pyglet

Check that your audio output device is working.

App looks different/wrong:

Use Tools > Cycle Theme to switch themes.

The app uses Segoe UI font (included with Windows 10/11).

GETTING HELP:

File issues at the project repository.

Include your Python version, OS, and the error message.